

make it more difficult to obtain credit and can result in higher interest rates and less favourable loan terms.

Worldwide, perhaps, the most popular credit rating agencies are **Moody's Investors Service (Moody's)** and **Standard & Poor's (S&P)**. In India, as of now, there are seven registered credit rating agencies. Their names are as follows:

- 1) Credit Rating Information Services of India Limited (CRISIL)
- 2) Investment Information and Credit Rating Agency of India (ICRA) Limited
- 3) Credit Analysis and Research (CARE) Limited
- 4) Acuite Ratings & Research,
- 5) Brickwork Ratings India Private Limited
- 6) India Ratings and Research Pvt. Ltd.
- 7) Infometrics Valuation and Rating Pvt. Ltd.

CRISIL is the oldest credit rating agency in India followed by ICRA and CARE. Let us see briefly their credit rating notations and explanations.

1) Credit Rating Information Services of India Limited (CRISIL)

Credit Rating Information Services of India Limited (CRISIL), established in 1987, is one of the oldest credit rating agencies in India. It operates in various countries like the UK, USA, Poland, China, Hong Kong, and Argentina, apart from India. It evaluates the creditworthiness of commercial entities based on their strengths, market reputation, market share and board. Its ratings range from CRISIL AAA to CRISIL D which are given below:

Table 12.1: Different categories of credit ratings given by CRISIL

Rating	Explanation	Credit Risk	Degree of safety
CRISIL AAA	Securities with this rating are considered to have the highest degree of safety regarding timely servicing of financial obligations. Such securities carry the lowest credit risk.	Lowest risk	Highest degree of safety
CRISIL AA	Securities with this rating are considered to have a high degree of safety	Very low risk	The high degree of safety

	regarding timely servicing of financial obligations. Such securities carry very low credit risk.		
CRISIL A	Securities with this rating are considered to have an adequate degree of safety regarding the timely servicing of financial obligations. Such securities carry low credit risk.	Low risk	Adequate degree of safety
CRISIL BBB	Securities with this rating are considered to have a moderate degree of safety regarding timely servicing of financial obligations. Such securities carry moderate credit risk.	Moderate risk	A moderate degree of safety
CRISIL BB	Securities with this rating are considered to have a moderate risk of default regarding timely servicing of financial obligations.	Moderate risk of default	-
CRISIL B	Securities with this rating are considered to have a high risk of default regarding the timely servicing of financial obligations.	High risk of default	-
CRISIL C	Securities with this rating are considered to have a very high risk of default regarding the timely servicing of financial obligations.	Very high risk of default	-
CRISIL D	Securities with this rating are in default or are expected to be in default soon.	Worst risk	-

Source: <https://www.crisilratings.com/en/home/our-business/ratings/credit-ratings-scale.html>

2) Investment Information and Credit Rating Agency of India (ICRA) Limited

The Investment Information and Credit Rating Agency of India, established in 1991, uses a rating system to give comprehensive credit ratings to corporates. ICRA is known for assigning corporate governance ratings, mutual funds ratings, structured finance ratings, performance ratings, etc. Credit ratings of ICRA range from ICRA AAA to ICRA D. Their explanations are the same as in CRISIL.

3) Credit Analysis and Research (CARE) Limited

Credit Analysis and Research Limited (CARE) was established in 1993. CARE rates bank loan ratings- short-term and long-term debt instruments. Credit ratings of CARE range from CARE AAA to CARE D. Their explanations are the same as in CRISIL.

Check Your Progress 4

- 1) What is a credit rating? How is credit rating useful for an investor?

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- 2) What are the credit rating codes of CRISIL, ICRA, and CARE?

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12.7 BOND MARKET IN INDIA

12.7.1 Bond Market — A Part of The Debt Market in India

The debt market, also known as the ‘fixed income market,’ includes instruments that offer fixed returns, with the bond market being a key component. In countries like India, debt markets have become a vital source of funds, serving as an alternative to traditional banking channels for financing. The significance of debt markets in an economy is underscored by their role in:

- Efficiently mobilizing and allocating resources.
- Funding the government’s developmental activities.

- c) Transmitting signals for the implementation of monetary policy.
- d) Managing liquidity in line with short-term and long-term objectives.
- e) Providing a credible alternative to banks as funding sources.

12.7.2 Regulators

As we have discussed in Unit 10 and Unit 11, Debt market transactions, including issuances and investments, fall under the jurisdiction of the following regulators:

- a) **RBI (Reserve Bank of India):** As the ‘money market’ regulator, the Reserve Bank of India regulates and facilitates government bonds and other securities on behalf of the government.
- b) **SEBI (Securities Exchange Board of India):** As the ‘securities market regulator,’ the Securities and Exchange Board of India regulates corporate and municipal bonds issued by entities that are listed or intend to be listed.

However, for money market instruments such as Commercial Paper, if the issuer intends to list them, the disclosures must comply with SEBI requirements.

12.7.3 The Growth of Indian Bond Market

India’s bond market is currently valued at US\$2.59 trillion, reflecting a 12.3% annual growth in INR terms from the previous year. The corporate bond segment, worth US\$567 billion, also saw a year-over-year growth of 12.6%. Despite this, the debt market is predominantly composed of government securities (G-secs). Specifically, the outstanding stock of government securities amounts to ₹1,68,85,262 crores (US\$2.02 trillion), while corporate bonds stand at ₹47,28,935 crores (US\$567 billion). Over the past decade (from 2013-2023), India’s bond market has grown at an impressive annual rate of 12.6% in INR terms, underscoring its expanding role in the country's financial ecosystem. The inclusion of Indian bonds in global indices by JP Morgan and Bloomberg has further highlighted Indian fixed income on the global stage. S&P Global Ratings has upgraded India’s credit rating outlook from ‘stable’ to ‘positive,’ citing policy consistency, significant economic reforms, and substantial infrastructure investments as key drivers of sustained long-term growth. With India’s GDP projected to reach US\$5 trillion in the coming years, much of this growth will be financed through credit provided to corporations via the bond markets. This trend is evident in Western economies, where corporate bond markets constitute a significant percentage of GDP (up to 50-65% in some cases). As India transitions from a developing to a developed economy, the sophistication and depth of its financial markets will be crucial for economic advancement. Households transferring their substantial savings to investments, particularly

into the safer asset class of corporate bonds, are expected to drive exponential growth in corporate bonds and support India's credit expansion.

Source: <https://www.indiabonds.com/bonduni/blogs/size-of-the-indian-bond-market-is-us-2-59-trillion/>

12.7.4 Classification of the Debt Instruments

In India, the classification (based on the issuer) of the debt or bond instruments is given below:

Market segment		Issuer	Instruments
Government Bonds or Government securities		Central, State Governments	Zero coupon bonds, Coupon bearing bonds, Treasury Bills, STRIPS, State Development Loans, Gold Bonds, Sovereign Green Bonds
Corporate Bonds	Public sector bonds	Government Agencies /Statutory Bodies	Government Guaranteed Bonds and Debentures, Municipal Bonds
		Public Sector Units (PSU)	Bonds, commercial papers
	Private sector bonds	Corporates	Bonds, commercial papers
		Banks	Certificate of Deposits, Bonds, Structured instruments and Perpetual Bonds
		Financial institutions	Certificate of Deposits, Bonds, Structured Instruments, Commercial Paper

Let us discuss these debt instruments briefly:

Government Securities: G-Secs are issued by the Reserve Bank of India on behalf of the Government of India. Normally the dated government securities have a period of 1 year to 30 years. These are sovereign instruments

generally bearing a fixed interest rate with interests payable semi-annually and principle as per schedule. For shorter term, RBI issues Treasury Bills which are discounted papers. At present T-Bills are issued for 91 days, 182 days & 364 days. G-Secs provide risk free (credit risk) return to investors.

STRIPS is the acronym for *Separate Trading of Registered Interest and Principal Securities*. Stripping is the process of separating a standard coupon-bearing bond into its individual coupon and principal components. For example, a 5-year coupon bearing (Government) bond (with half yearly coupon payment) can be stripped into 10 coupons and one principal instrument, all of which thenceforth would become zero coupon bonds.

Sovereign Green Bonds : The budget division of the ministry submits projects approved by GFWC for bond issuance through the Reserve Bank of India (RBI). These bonds, known as sovereign green bonds, are issued by the central bank to finance green initiatives. The Ministry of Finance maintains a green register containing bond details to monitor these bonds. Notably, the ministry releases an annual report on these projects, detailing bond issuances, approved projects, allocated expenditures, current statuses, and potential impacts on reducing carbon intensity.

Certificate of Deposit (CD) : As discussed in Unit 10, CDs are negotiable money market instruments issued in demat form or as a Usance Promissory Notes. CDs issued by banks should have a maturity of not less than seven days and not more than one year. Financial Institutions are allowed to issue CDs for a period between 1 year and up to 3 years. CDs normally give a higher return than Bank term deposit. CDs are rated by approved rating agencies (e.g. CARE, ICRA, CRISIL, FITCH) which considerably enhances their tradability in secondary market. CDs are issued in denominations of Rs. 1 Lac and in the multiples of Rs. 1 Lac thereafter.

Commercial Papers : As we discussed in Unit 10, a CP is a short-term security (7 days to 365 days) issued by a corporate entity (other than a bank), at a discount to the face value. One can invest in CPs starting from a minimum of 5 lacs (face value) and multiples thereof. CPs are rated by approved rating agencies (e.g. CARE, ICRA, CRISIL, FITCH). CPs normally give a higher return than fixed deposits & CDs. We deal in investment grade CPs only. CPs can be traded in the secondary market, depending upon demand. An element of credit risk is attached to CPs.

12.8 LET US SUM UP

Bond is a way of raising long-term loans. A bond is traded in a market known as a bond market which is classified as a primary market and a secondary market. The primary market deals in new bonds while the secondary market deals in second-hand bonds. The market forces of demand and supply determine the market price (or just price) of the bond, but its intrinsic value is determined by calculating the present value of all the coupons and redemption value. There are three types of bond yields: Yield to Maturity

(YTM), Yield to Call (YTC), and Current Yield. These yields are generally quoted by the issuers to attract investors. Bonds are of various types e.g., government bonds, corporate bonds, etc. Bonds carry default risk too; therefore, credit rating agencies rate the bonds or issuer companies/governments so that investors can know the creditworthiness of the issuer.

12.9 KEY WORDS

Bonds	: Bonds are long-term debt instruments.
Coupon	: It is the interest amount on a bond.
Bond market	: A place where bonds are traded.
Value of bond	: The value of a bond means the intrinsic value of a bond. It is calculated as the present value of all the coupons and the redemption value of it.
Price of bond	: It is the market price of a bond which is determined by the rules of demand for and supply of bond.
Yield to Maturity (YTM)	: It is the rate of return on a bond if the bond is held till its maturity.
Yield to Call (YTC)	: In the case of a callable bond, it is the rate of return on a bond if the bond is held till the time of the call.
Current yield	: It is a rate of return in terms of interest per ₹ 1 of investment.
Credit rating	: Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full.
Credit rating agency	: It is the agency which performs credit rating. CRISIL, CARE, and ICRA are examples of credit rating agencies in India.

12.10 SOME USEFUL BOOKS

Brigham, E., & Ehrhardt, M. (2017). *Financial Management* (12th ed.). Thomson South Western.

Horne, J. V., & Wachowicz, J. (2008). *Fundamentals of Financial Management* (13th ed.). Prentices Hall.

Kishore, R. M. (2010). *Strategic Financial Management* (2nd ed.). Taxmann Publications Pvt. Ltd.

Pandey, I. M. (2009). *Financial Management* (9th ed.). Vikas Publishing House Pvt. Ltd.

Luenberger, D. G. (1998). *Investment Science*. Oxford University Press.

12.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A bond is a long-term debt instrument in which the issuer of the bond is bound to pay a fixed or floating interest rate and repay the principal amount after a fixed period. Thus, a bond is a way of taking long-term funds.
- 2) **Bonds Based on Issuer:** Government bonds, Corporate bonds, Municipal bonds, Foreign bonds. **Bonds Based on Coupon:** Zero coupon bonds, Coupon bonds. **Bonds based on maturity:** Perpetual bonds, non-perpetual bonds. **Bonds based on option:** Puttable bond, Callable bond. **Bonds based on convertibility:** Convertible bonds, non-convertible bonds.

Check Your Progress 2

- 1) Refer to Section 12.4
- 2) A bond with a face value of ₹ 5,000 matures at the end of 5 years, the rate of interest on the bond being 15% p.a. paid annually. Find the value of the bond so as to yield a return of 10% p.a. Further, if the bond is trading at ₹ 6,000, then is it worth to buy it? Assume that the bond will be redeemed at par.

Step 1: Summarise the information

Face value = ₹ 5,000

Coupon rate = 8%

Coupon = Coupon rate $\times$ face value = 15% $\times$ ₹ 5,000 = ₹ 750

Discount rate = i = 10% = 0.10

Maturity of bond = n = 5 years

Redemption value = RV = Face value = ₹ 5,000

Step 2: Apply the following formula

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \frac{C_3}{(1+i)^3} + \cdots + \frac{C_n}{(1+i)^n} \right] + \frac{RV}{(1+i)^n}$$

$$= \left[\frac{750}{(1+0.10)^1} + \frac{750}{(1+0.10)^2} + \frac{750}{(1+0.10)^3} + \frac{750}{(1+0.10)^4} + \frac{750}{(1+0.10)^5} \right] + \frac{5,000}{(1+0.10)^5}$$

$$\begin{aligned}
&= \left[\frac{750}{(1.10)^1} + \frac{750}{(1.10)^2} + \frac{750}{(1.10)^3} + \frac{750}{(1.10)^4} + \frac{750}{(1.10)^5} \right] + \frac{5,000}{(1.10)^5} \\
&= [750 \times 0.9090 + 750 \times 0.8264 + 750 \times 0.7513 + 750 \times 0.6830 + 750 \times 0.6209 +] \\
&\quad + 5,000 \times 0.6209 \\
&= [681.8182 + 619.8347 + 563.4861 + 512.2601 + 465.691] + 3104.607 \\
&= 5947.697
\end{aligned}$$

Since the bond is being traded at ₹ 6,000 and the intrinsic value is ₹ 5947.697, therefore, the bond is undervalued and it is worth to buy it.

Check Your Progress 3

- 1) Refer to Sub-section 12.5.1
- 2) Calculate the yield to maturity from the following information. Coupon rate = 7%, face value = ₹ 500, price of bond = ₹ 350, maturity period = 10 years.

$$\text{Approximate YTM} = \frac{C + (F - P)/n}{(F + P)/2}$$

$$\text{Approximate YTM} = \frac{35 + (500 - 350)/10}{(500 + 350)/2}$$

$$= \frac{35 + 15}{425}$$

$$= \frac{50}{425}$$

$$= 0.1176$$

$$= 11.76\%$$

Note: The above formula is approximation only. If you are interested to find the exact YTM, then the trial-and-error method will give us 12.396% as YTM. In order to understand the trial-and-error method, the students are suggested to read to calculate IRR by referring any book on Financial Management.

- 3) Refer to Sub-section 12.5.2 and Sub-section 12.5.3

Check Your Progress 4

- 1) Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full.
- 2) Refer to Table 12.1

UNIT 13 DERIVATIVES

Structure

- 13.0 Objectives
- 13.1 Introduction
- 13.2 Meaning of Derivatives
- 13.3 Characteristics of Derivatives
- 13.4 A Brief History of Derivatives in India
- 13.5 Types of Derivatives
- 13.6 Futures and Forwards
 - 13.6.1 Meanings of Futures and Forwards
 - 13.6.2 Differences between Futures and Forwards
- 13.7 Options
 - 13.7.1 Call Option
 - 13.7.2 Put Option
- 13.8 Swaps
- 13.9 Let Us Sum Up
- 13.10 Key Words
- 13.11 Some Useful Books and References
- 13.12 Answers/Hints to Check Your Progress Exercises

13.0 OBJECTIVES

After going through this unit, you will be able to:

- state the meaning of derivatives with their characteristics;
- explain the main types of derivatives;
- discuss the pros and cons of derivatives; and
- distinguish between futures and forward contract.

13.1 INTRODUCTION

In Unit 10 and 11, you have understood various types of financial assets e.g., equity shares, stock, debentures, fixed deposits, and so on. All these assets share a common trait that their values are not linked to the value of some other asset. However, there are some unique financial assets known as derivatives which derive their values from some other assets known as underlying assets. These underlying assets may be shares, stocks, debentures,